



Product Requirements Document

Stoma Measuring & Automated Wafer Cutting System

v0.2 — Investor Demo Build

Client	Cole Fields (Founder, Owner; TACKLE Medical)
Prepared by	DryDock — Aaron Gopp (Lead Engineer), Blake Jackson (Project Manager)
Source	Technical review & demo call — "Cole <> DryDock", Wednesday 19 August 2026, 40 minutes (Fathom recording)
Date	20 August 2026
Version	0.2 — supersedes v0.1. Scope reduced to the investor-demo build.
Status	Pre-contract. Pricing, timeline, invoice and service agreement to follow by Friday 21 August.

0. What Changed in This Version

v0.1 scoped a ground-up product build: patient app, backend, database, HIPAA compliance layer, processing automation, device connectivity and a validation build, phased across six stages. The 19 August call reset the target.

The near-term objective is now a single thing — an investor-demonstrable end-to-end build. Compliance was explicitly deferred by agreement. Everything not required for the demo has been removed from scope and is listed in §9 so nothing is lost, only postponed.

This document reflects only what was discussed and agreed on 19 August. Items marked [TBC] are open. Items marked [Assumption] are DryDock's working interpretation.

1. Objective

1.1 Primary objective

A working demonstration Cole can put in front of investors. In his own framing on the call: a patient-facing app showing what the patient would see, plus a back-end view that outputs "here is the measurement we obtained, here is the outline" — sent to the cutting device, with a cut wafer produced on the spot.

The claim the demo has to support: this works, end to end, within acceptable tolerances.

1.2 Secondary objective

A verification output Cole can feed into his own engineering and design control documents — evidence that the next prototype meets the stated design inputs. Doubles as investor material ("we have run N tests across N unique stomas at an average margin of X mm").

1.3 Explicit non-objectives

- Production readiness or live patient use.
- HIPAA compliance implementation.
- Regulatory submission.

Rationale recorded on the call: at this funding stage, an immediately user-ready product is rarely what unlocks investment — funders typically respond by asking for users. Building to that standard now costs tooling spend, build time and delay without improving the funding position.

2. Commercial & Timeline Context

Budget	Cole is targeting the ~\$5,000 range from disposable funds available before the raise. DryDock confirmed the demo scope sits within comfortable range of that, subject to codebase review. No firm number was quoted on the call.
Pricing trigger	Pricing follows Aaron's review of the codebase against Cole's design control document. Deliberately not quoted before review.
Timeline	No hard deadline. Cole is in two competitions; the earlier final is in late October, explicitly not set as a delivery deadline. Left open-ended by agreement.
NDA	Executed. Cole signed and returned via DocuSign; DryDock countersigned.
Service agreement	To follow. Contractor terms, no benefits, DryDock retains autonomy on tooling, Cole owns 100% of everything built, NDA baked in. Kept intentionally general on engagement detail so scope changes do not require re-papering.
What changes later	Cost scales at the point compliance and post-demo development begin. Compliance is expected to be a long project in its own right.

3. Current State — As Demonstrated 19 August

3.1 The pipeline as it stands

- Swift application, written with Cursor (base Cursor app), running locally on Cole's Mac. Not in version control — the codebase exists only on his machine.
- Video is imported; keyframes are extracted at a current rate of one frame every 0.1 seconds.
- Frames feed an Apple ARKit photogrammetry session, which reconstructs a 3D mesh.
- Scale is derived from an ArUco fiducial marker of known size. The app orbits a virtual camera around the mesh, captures a snapshot where the marker is detected, pinpoints its corners, and uses the known dimensions to establish real-world scale. A marker is dropped onto the mesh to confirm successful detection.
- Cross-section: Cole manually fine-tunes the X and Z axes to place the slice at the base of the stoma.
- The base perimeter is extracted and the diameter derived. On the demo model the app read 33 mm; a caliper check at one point on the physical model read 31.4 mm.



- A 3 mm grace ring is applied around the outline, per wafer manufacturer recommendation, allowing the stoma to expand and contract.
- The result is exported to a G-code plotter.
- Current processing time: approximately two to two and a half minutes from frame extraction to mesh. Cole's own assessment: "kind of long" — acceptable for now, but he would rather it were much shorter.

3.2 Why photogrammetry rather than LiDAR

Photogrammetry carries no inherent scale, which is why the ArUco marker is needed. LiDAR would supply scale directly, but is only available on Pro-model iPhones and is not a standard Android feature. Pure photo/video capture was chosen deliberately to keep non-LiDAR devices in the addressable population. This constraint carries forward.

3.3 Known gaps

- Base detection is manual. Two specific problems were named on the call: (a) the system has no notion of "up," so orienting the positive axis perpendicular to the base of the stoma is unsolved — and that orientation determines the plane the slice translates on; (b) determining which Y value to take the cross-section at.
- On an irregular stoma the diameter differs depending on where it is measured. There is no defined reference point on the physical stoma for where truth is measured, so app-versus-caliper comparison is not yet like-for-like.
- Everything demonstrated is the back-end view. There is no patient-facing application.
- Mac / Swift only, running locally.
- No automatic connection to the cutting device.

4. Design Constraints

Taken from Cole's design control document, which was shared by email during the call. DryDock to reconcile this PRD against that document on review.

Constraint	Value	Source
Dimensional tolerance	±1 mm on the stoma base measurement	Cole — design input file
Cycle time	Cut complete within 2 minutes	Blake — design input
Aperture offset	3 mm grace ring around the extracted outline	Wafer manufacturer recommendation
User capability	No fine-tuning, no parameter selection, no technical judgement. 48% of the patient population is over the age of 60.	Cole
Device dependency	Must not require LiDAR	Cole

The design control document covers user needs, functional requirements, safety requirements, performance and physical requirements, software, data, regulatory, QMS, and verification & validation planning.

5. Scope — Investor Demo Build

5.1 A. Architecture migration

- Lift the processing maths and rendering out of Swift into a platform-agnostic backend service. Recommended approach: port to a Python service hosted on DigitalOcean, with the front end — iOS, Android, React, or otherwise — calling back to that process.
- Database: Supabase. Starts at \$3–10/month and scales to enterprise, including a signed BAA and HIPAA compliance when that becomes necessary. Chosen so the compliance phase is an upgrade rather than a rebuild.
- Hard requirement stated by Cole and agreed: every tool and platform selected must scale without forcing re-architecture later because of an early shortcut.

5.2 B. Processing automation

- Automatic base detection, replacing the manual X/Z slice selection — solving both axis orientation and slice height.
- Aaron's assessment on the call: doable, will take fine-tuning like anything else, and nothing in the pipeline reads as a blocker.
- Retain the 3 mm offset, configurable rather than hard-coded.
- Reduce cycle time toward the 2-minute target — principally by establishing how few keyframes still yield a usable mesh, and by moving processing off the local machine.

5.3 C. Patient-facing application (demo grade)

- A front end representing what the patient would actually see. The patient's entire job: record a video, then collect the cut wafer from the machine.
- No manual tuning, no mesh views, no exposed processing parameters.
- Quality sufficient to present to investors without caveating unfinished areas.

5.4 D. Device connectivity

- Generate a standard G-code file and transmit it to the cutter automatically.
- Target controller: GRBL, the G-code motion controller expected on the custom machine. DryDock's job is to connect to GRBL and send it what it needs to cut.
- Transport: Bluetooth or Wi-Fi. Treated as low risk — the integration is largely the same either way.
- Firmware, motion control and G-code interpretation on the machine remain with the manufacturer. DryDock is not building Arduinos or the machine-side G-code execution.

5.5 E. Verification & test-log module

- Compare each derived outline against the configured ± 1 mm tolerance and report the deviation.
- Maintain a running log of test runs, exportable, with per-run deviation and an aggregate margin.

- Two uses: evidence for Cole's verification testing and design control documentation, and an investor-facing credibility artefact alongside the live demo.

5.6 Out of scope

- HIPAA compliance implementation. Deferred by explicit agreement — the stack is architected to accept compliance at any point, with no rebuild, but no compliance work is done in this phase.
- Patient records, identity management, de-identification, retention policy, role-based access, audit logging.
- Hardware design and manufacture of the cutting device — Remedy Medical Manufacturing.
- CNC firmware, motion control, and machine-side G-code interpretation.
- Clinical, regulatory and QMS ownership. Cole's. DryDock supplies the software and the test-evidence output.
- Clinician portal, EHR integration, reimbursement workflow.

6. Functional Requirements

6.1 Capture & processing

ID	Requirement	Priority
FR-01	The system shall accept a recorded video as input and extract keyframes for photogrammetric reconstruction.	Must
FR-02	The system shall reconstruct a 3D mesh of the stoma from the extracted frames.	Must
FR-03	The system shall derive real-world scale from a known-size ArUco marker, without requiring LiDAR.	Must
FR-04	The system shall determine the orientation of the stoma base automatically, establishing the axis perpendicular to the base with no user input.	Must
FR-05	The system shall select the cross-section height automatically. No manual X/Z fine-tuning by any user in the patient flow.	Must
FR-06	The system shall extract the base perimeter and derive the stoma outline and diameter.	Must
FR-07	The system shall apply a 3 mm offset to the extracted perimeter. The value shall be configurable rather than hard-coded.	Must
FR-08	The system shall generate a standard G-code file from the offset outline.	Must
FR-09	Derived measurements shall fall within ± 1 mm of the physical stoma base.	Must
FR-10	The system shall define and consistently apply a reference point for where the base diameter is measured, so app output and physical measurement are comparable. [TBC] — raised on the call as unresolved.	Should

ID	Requirement	Priority
FR-11	End-to-end measure-to-cut-complete shall target ≤ 2 minutes. Current pipeline is ~2–2.5 minutes to mesh alone.	Should

6.2 Patient application (demo)

ID	Requirement	Priority
FR-12	The app shall present a patient-facing flow: record video, wait, collect the cut wafer.	Must
FR-13	The patient shall not be exposed to mesh views, slice controls, or any processing parameter.	Must
FR-14	The app shall display the captured measurement and resulting outline in a form suitable for live demonstration to investors.	Must

6.3 Device connectivity

ID	Requirement	Priority
FR-15	The system shall transmit the generated G-code to the cutting device automatically, with no manual file handling.	Must
FR-16	Transport shall be wireless — Bluetooth or Wi-Fi, selected once the hardware is specified.	Must
FR-17	The system shall target a GRBL-based G-code motion controller. [Assumption] — to be confirmed with Remedy Medical Manufacturing.	Must
FR-18	Output shall be a standard G-code file. Whether the machine expects tool-specific parameters is [TBC] — an example G-code file is needed from the manufacturer.	Must

6.4 Verification module

ID	Requirement	Priority
FR-19	The system shall compare each derived outline against the configured tolerance and report the deviation.	Must
FR-20	The system shall maintain a running, exportable log of test runs recording per-run deviation and aggregate margin.	Must
FR-21	Log output shall be in a form usable as evidence in Cole's design control and verification documentation.	Must

7. Non-Functional Requirements

ID	Requirement	Priority
NFR-01	The backend shall be platform-agnostic and independent of macOS and Swift, so front ends for iOS, Android or web can be built against the same service.	Must
NFR-02	Every platform and tool selected shall scale from demo to production without re-architecture.	Must
NFR-03	The stack shall be capable of HIPAA-compliant configuration at a later date — for example via a signed Supabase BAA — without rebuilding.	Must
NFR-04	Measuring shall not depend on LiDAR, keeping non-Pro iPhones and Android devices in scope.	Must
NFR-05	The patient flow shall require no fine-tuning, parameter selection or technical judgement. 48% of the patient population is over 60.	Must
NFR-06	Visual and interaction quality shall be sufficient to present to investors without caveat.	Should
NFR-07	Compliance controls are explicitly not implemented in this phase. This is a deliberate exclusion, not an omission.	Must

8. Assumptions

- Cole's existing measuring, mesh, scale-calibration, perimeter extraction and G-code logic is a usable foundation and does not need re-deriving from first principles.
- The cutting machine will run a GRBL-based G-code controller; sending it a standard G-code file is sufficient to produce a cut.
- Hardware development stays with Remedy Medical Manufacturing and is not on DryDock's critical path. Device connectivity can be developed against a bench target.
- The demo target is investor demonstration and Cole's own verification testing, not patient use.
- Cole owns the regulatory pathway, clinical requirements and QMS. DryDock supplies software and test-evidence output.

9. Deferred From v0.1

Not cancelled — postponed. Recorded here so scope can be restored deliberately rather than by drift.

Item	Why deferred
HIPAA compliance layer and third-party compliance monitoring	Agreed on the call to push down the road. No funding benefit at demo stage; costs tooling spend, build time and delay. Architected for, not built.
Patient identity, de-identification, measurement history retention, role-based access, audit logging	Dependent on the compliance phase.
Guided real-time capture coaching (distance, lighting, coverage prompts)	Not discussed on this call. Carried forward as a candidate, not committed.
Separate developer / validation build	Superseded by the verification & test-log module (\$5.5), which serves the same purpose inside the main build.
Clinician interface, EHR integration, reimbursement	Not discussed. Unchanged from v0.1.
Human factors trial support and demo-hardening phase	Not discussed on this call.
Six-phase build plan (Phases 0–5)	Superseded by a single demo-scope engagement.

10. Open Questions

- Where on the physical stoma is the base diameter measured for verification purposes? Without a fixed reference point, app-versus-caliper comparison is not like-for-like.
- Which wireless transport does the machine support — Bluetooth, Wi-Fi, or both? Awaiting Remedy, currently occupied with an FDA audit.
- Can the manufacturer supply an example G-code file? Specifically, does the machine expect tool-specific parameters, or is standard G-code sufficient? Aaron's view: tooling specifics ought to sit on the CNC side, not the software side.
- Which platform hosts the demo front end — iOS, Android, or web?

11. Risks

Risk	Impact	Mitigation
Automatic base detection — axis orientation and slice height — proves harder than expected. This is the core unsolved technical problem.	High	Aaron assessed it as solvable with fine-tuning. Prototype early against Cole's existing test models; the verification module

Risk	Impact	Mitigation
		quantifies progress against the ± 1 mm target continuously.
The manufacturer is unresponsive (preparing for an FDA audit), leaving machine specification and connectivity unknown.	Medium	Build device connectivity against a bench GRBL target. Keep the software path independent of hardware readiness.
± 1 mm tolerance is not achievable from video photogrammetry.	Medium	Cole's assessment: close enough that fine-tuning gets there, and the standard of care is a stencil and scissors, so the improvement bar is low. The verification log surfaces this early rather than at trial.
Cycle time exceeds the 2-minute design input.	Low–Medium	Keyframe optimisation and offloading processing from the local machine to the hosted service.
Scope drifts toward compliance before funding lands.	Medium	Explicitly excluded from this phase; the stack is built to accept it later without rebuild. Revisit at funding.

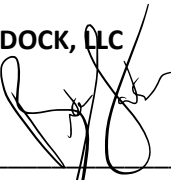
Document Acknowledgment

This Product Requirements Document reflects the shared understanding of Provider and Client regarding the project's scope, assumptions, and expectations as of the date signed below. Both parties confirm they have reviewed this document and agree it accurately represents what will be delivered.

Any changes to this scope after signing will be handled via a written change order and may affect timeline and/or fees as outlined in the Service Agreement.

Agreed and accepted:

DRYDOCK, LLC

By: 

Name: Blake Jackson,
Co-Founder

Date: 8/21/26

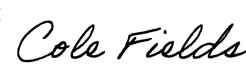
DRYDOCK, LLC

By: 

Name: Aaron Gopp,
Co-Founder

Date: 8/21/2026

TACKLE MEDICAL, LLC

By: 

Name: Cole Fields
Founder

Date: 08/21/26